

TrES-5b

R-1816

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220721 · created 2026-08-03T04:07:14+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459781.7869 +/- 0.0035 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.047
Transit depth (Rp/Rs)^2	3.4 +/- 1.73 [%]
Orbital Inclination (inc)	82.37 +/- 2.57
Airmass coefficient 1 (a1)	0.501 +/- 0.03
Airmass coefficient 2 (a2)	0.58 +/- 0.15
Scatter in the residuals of the lightcurve fit is	5.82 %
Best Comparison Star	#2 - [595, 474]
Optimal Method	PSF photometry
Transit Duration (day)	0.06 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.184 ± 0.047 vs 0.142 (deviation 0.86σ)
Duration fit vs published	0.06 d vs 0.0758 d (deviation 0.66σ)
Mid-time O-C	-3.1 min
Depth SNR	3.8
Residual scatter	5.82 %

Light curve

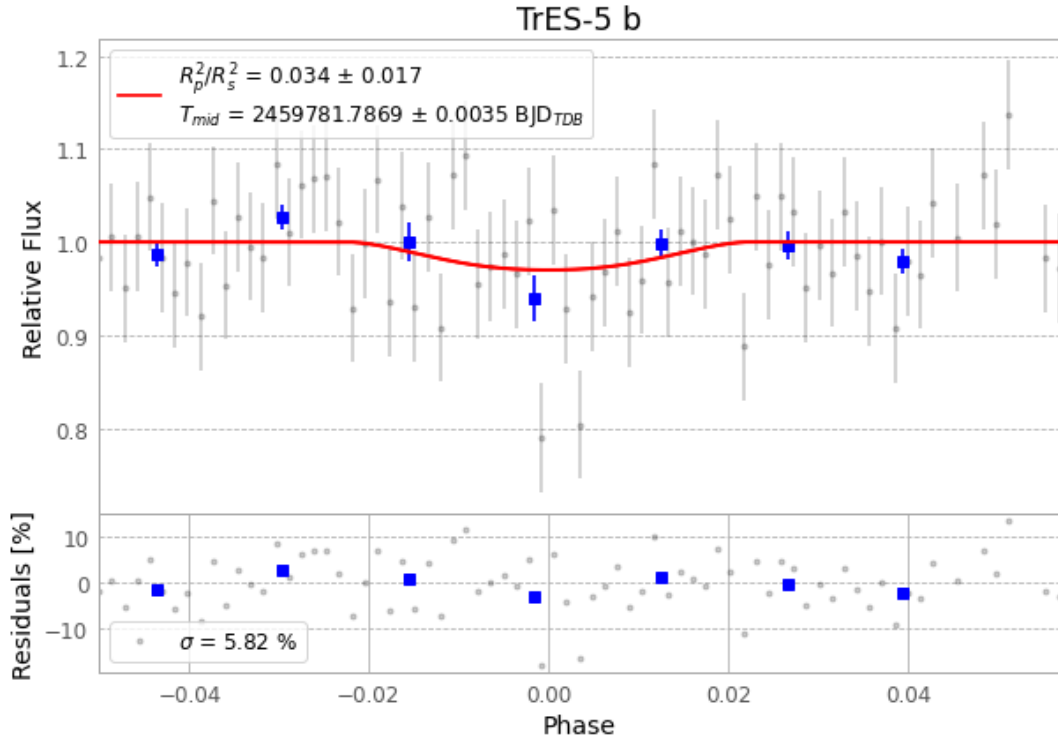


Figure 1 — FinalLightCurve_TrES-5 b_2022-07-20.png (SHA-256 4e711a35a3af7e6e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [498, 246], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 52348df2038a9531...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), TRES-5220721050313.FITS → TRES-5220721085113.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220721.log (327809a315c4...), FinalLightCurve_TrES-5 b_2022-07-20.png (4e711a35a3af...), FinalLightCurve_TrES-5 b_2022-07-20.csv (db504375b40c...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:07Z.